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Prevention of deep venous thrombosis and pulmonary embolism following stroke: a systematic review of published articles

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We performed a systematic review of the literature on venous thromboembolism (VTE) prophylaxis following cerebral infarct (CI) and haemorrhagic stroke. MEDLINE, Cochrane, LILACS and SciELO databases were scanned, and the Abstracts from Brazilian, American and European Neurology and Stroke Congresses were scrutinized for clinical trials. Moreover, the reference lists of articles and reviews were searched. A pooled analysis of two large studies with aspirin was made. Both unfractionated heparin and low molecular weight heparins/heparinoids (LMWH) are partially effective for VTE prophylaxis after CI, and should be routinely used in patients with motor deficit and reduced mobility and no contraindications. Reduction of deep venous thrombosis is better established than the effect over pulmonary embolism or mortality. Some evidence points to a greater efficacy of LMWH. The available evidence does not support the use of mechanical methods or dextran. Aspirin may have a mild protective effect. Low-dose Warfarin might be useful in the rehabilitation setting. Strict recommendations cannot be made in patients with haemorrhagic stroke but intermittent pneumatic compression merits further study. There are important limitations of current VTE preventive strategies following stroke. Additional studies on the combination of methods after CI and of low doses of anticoagulants following cerebral haemorrhage are urgently needed.

Introduction

Immobility is the most important risk factor for venous thromboembolism (VTE) [1]. It is hence not surprising that VTE constitutes one of the most important complications of all types of stroke. Deep venous thrombosis (DVT) is frequently detected (15–80%) in prospective studies using fibrinogen uptake or venous Duplex-scan in patients with cerebral infarcts (CI) without prophylaxis. Although asymptomatic DVT may have long-term consequences such as the post-phlebitic syndrome and a predisposition for future recurrent VTE, its clinical relevance in the acute setting is doubtful, especially when confined to the calf veins. Most studies show that pulmonary embolism (PE) seems to be much more common in patients with proximal and symptomatic DVT [1]. Studies evaluating only clinically evident DVT report consistently lower rates – between 0.2–10%. However, many of these studies did not evaluate suspected VTE events prospectively and are probably clear underestimates of their true incidence. The same restriction applies to the reported 0.2–5.7% frequency of PE in acute stroke patients [2–6].

The best strategy to prevent DVT and PE after stroke is not well established. The use of antithrombotic drugs seems to reduce the frequency of DVT and PE but is counterbalanced by an increased risk of extra and intracranial bleeding. Aspirin should be routinely used soon after CI but its role as an adjunctive method to prevent VTE is not clear. Mechanical methods are frequently used in patients considered to be at high risk of bleeding but the scientific support for their recommendation is at least fragile. Recent guidelines and reviews recommend the use of low doses of unfractionated heparin (UFH) or of low molecular weight heparins (LMWH) but also state that this practice is associated with some risk of systemic or intracranial bleeding, which may be catastrophic [7–9].

The purpose of this study is to systematically review clinical studies of VTE prophylaxis in stroke patients. The emphasis is in acute CI but we also analysed studies addressing the rehabilitation phase of the disease and patients with acute cerebral haemorrhage (CH) or subarachnoid haemorrhage. An effort

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is made to distinguish the effects of the interventions on different outcomes such as low-risk calf vein thrombosis and higher-risk events such as proximal DVT and PE.

Methods

We performed a systematic review of the literature on thromboprophylaxis in stroke. Articles were included only if they evaluated one or more active treatments versus placebo, no intervention or other treatment modalities in adult (minimum 18 years) stroke patients. Clinical trials and systematic reviews were searched for in MEDLINE, COCHRANE, LILACS and SciELO databases. The search strategy was developed for MEDLINE (available from the authors upon request) and adapted for the other databases. Studies presented in Neurological and Cerebrovascular Diseases Congresses and only published in abstract form from 1998 to August 2004 were also searched for in the following journals: *Stroke*, *Cerebrovascular Diseases*, *Circulation*, *Journal of Neurology*, *Neurology*, *Arquivos de Neuro-psiquiatria*. The reference lists of all included articles and other pertinent published articles and reviews were also scrutinized in search for additional clinical studies of interest.

Two authors (CA and GRF) independently reviewed all titles and abstracts of the initially found studies for further assessment. Abstracts selected by any one of the authors were by necessity thoroughly analysed in the complete publication.

Articles were rated according to perceived scientific value following Cochrane Criteria [10] as follows:

A – Low risk of bias: Large (100 patients or more) randomized controlled clinical trials including systematic DVT evaluation in acute stroke patients and with all quality criteria met, i.e. appropriate randomization and blinding processes clearly described and $<20\%$ loss to follow-up; also, systematic reviews that included at least one trial with these characteristics.

B – Moderate risk of bias: Clinical trials of lower scientific value including small number (<100) of patients or many losses for follow-up.

C – High risk of bias: Studies with no systematic search for DVT, using historical controls, non-randomized, not presenting results by groups or with other major problems.

The efficacy of four main types of intervention was evaluated:

1 mechanical devices including elastic socks and intermittent pneumatic compression (IPC) devices;

2 parenteral anticoagulants: all as a group; UFH; LMWH and heparinoids; comparisons between LMWH/heparinoids and UFH;

3 aspirin: versus placebo or no treatment; or versus LMWH; we performed a meta-analysis of two large clinical studies comparing aspirin use and no treatment in CI [4,5];

4 other drugs.

Results

Number and quality of studies

We were able to find and review 29 clinical studies and six systematic reviews fulfilling our pre-specified criteria for inclusion (Table 1). An additional retrospective study evaluating mechanical devices in 151 patients with CH was later included as it is the first publication addressing the subject [11].

Most studies were flawed by major scientific drawbacks including small number of patients and absence of rigid criteria for diagnosing DVT/PE but there were six high quality studies including three large randomized trials with prospective evaluation of DVT and three systematic reviews. Twenty-five studies evaluated UFH or LMWH/heparinoids against each other, placebo or aspirin in the acute phase of CI. Twelve studies evaluated mechanical devices or other antithrombotics (aspirin, dextran or Warfarin) in the acute or rehabilitation phase of stroke or low-dose anticoagulation in CH.

Treatments and procedures

Mechanical devices

One high-quality trial and three clinical trials of low quality were found [11–14]. Their results are conflicting: a recent trial comparing elastic stockings plus IPC versus stockings alone suggested a marked risk reduction of asymptomatic DVT in the first group [11]. Both proximal (no case in the IPC group versus three cases in controls) and distal DVT were reduced. Although the study included patients with CH of vascular and traumatic origin (percentages not given), the results were similar in patients undergoing or not haematoma evacuation and after controlling for a number of factors including age.

In patients with CI, only the non-randomized trial evaluating UFH plus IPC suggested benefit [13]. A systematic review with meta-analysis only included the two small randomized studies and was also negative [15]. Aspirin plus IPC seemed ineffective in a single prospective observational study [16].

In summary, the available evidence does not support the routine use of elastic stockings and IPC devices in CI. Their efficacy as an adjunctive measure in CI patients receiving anticoagulants is not proved, having

Table 1 Trials and reviews: prevention of venous thromboembolism prophylaxis after stroke

Reference number	Patients	Trial design	DVT in controls – *n* (%)	DVT with intervention – *n* (%)	RR/OR[a] (95% confidence interval)	NNT[b]	Study quality
Mechanical methods – elastic stockings and intermittent pneumatic compression							
Muir [14]	97, CI <7 days	RCT, Complete; ultrasound	7/32 (22)	7/65 (11)	RR 0.43 (0.14–1.36); no effect over proximal DVT or PE	9 (5 to_13)	B
Prasad [12]	26, stroke < 7 days	RCT, Complete; fibrinogen uptake	6/13 (46)	6/13 (46)	RR 1 (0.44–2.29); no effect over proximal DVT or PE	0 (3 to_3)	B
Mazzone [15]	123, stroke < 7 days	SRMA, Complete	13/45 (26)	13/57 (16)	OR 0.59 (0.24–1.48)	16 (4 to _9)	B
Kamran [13]	681, CI (<?)	RCT, Complete; clinical DVP & PE	Heparin only 5000 IU b.i.d.: DVT 23/249 (9), 6 PE (2.4)	Heparin 5000 IU b.i.d. + IPC: DVT 1/432 (0.2), 0 PE	RR 0.03 (0 –0.18)	11 (8–19)	C
Lacut *et al.* [11]	151, CH (<?)	RCT, complete; DVP & PE	Elastic stockings: symptomatic DVT or PE 1/77, asymptomatic DVT 11/69 (15.9)	Stockings plus IPC: symptomatic DVT or PE 1/74, asymptomatic DVT 3/64 (4.7)	Asymptomatic DVT (D10): RR 0.29 (0.08–1.00)	9 (5–84)	A
Heparin versus controls							
McCarthy [20]	305, CI < 2 days	RCT, Complete; fibrinogen uptake & autopsy	DVT 117/161 (73)	5000 IU t.i.d.: DVT 32/144 (22)	RR DVT 0.31 (0.22–0.42); PE RR 0.65 (0.45–0.96)	2 (2–2)	A
Sandercock [2]	9 studies, 823, CI & CH; PE: 3 studies, 238, CI	SRMA, Complete	DVT 216/404 (53.4); PE 6/106 (5.7)	DVT 76/419 (18.1); PE 3/132 (2.3)	RR 0.34 (0.27–0.42) RR PE 0.40 (0.10–1.57)	3 (2–3); PE 30 (12 to 59)	A
McCarthy [17]	32, stroke < 2 days	RCT, Complete; fibrinogen uptake	12/16 (75)	5000 IU t.i.d. 2/14 (12)	RR 0.19 (0.05–0.71)	2 (1–3)	C
Czechanowski [18]	81 (107), CI (<?)	RCT, Complete; fibrinogen uptake	DVT 23/41 (56); suprapatelar – 4, bilateral - 6	Heparin (5000 IU b.i.d.) + Dihydroergot DVT 11/40 (27.5); suprapatelar 3, bilateral - 1	RR 0.49 (0.28–0.87)	3 (2–12)	C
Marti-Vilalta [19]	124, CI or CH < 7 days	RCT, Complete; clinical DVT & PE	6/62 (10)	2 regimens: 5000 IU b.i.d or t.i.d. 0/62 (0)	RR 0	10 (6–43)	C
Dickman [21]	46, CH < 1 day	RCT, Complete; clinical	Heparin D10 5000 IU t.i.d 2/23 (9); PE 5	Heparin D4 5000 IU t.i.d. 1/23 (4); PE 10	RR 0.5 (0.05–5.14); n.s. also for PE	23 (5 to _10)	C

Table 1 (Continued)

Reference number	Patients	Trial design	DVT in controls – *n* (%)	DVT with intervention – *n* (%)	RR/OR[a] (95% confidence interval)	NNT[b]	Study quality
IST [5]	19 435, stroke < 2 days	RCT, Complete; clinical PE	Aspirin or no prophylaxis 81/9717 (0.8)	2 regimens: 5000 or 12 500 IU b.i.d. 53/9716 (0.5)	RR 0.65 (0.46–0.92)	347 (192–1804) BUT significant ↑ extracranial bleeding & CH (both $P < 0.0001$)	C
Wasay [22]	458, CH (<?)	Observational, Abstract; clinical DVT & PE	1/258 (0.3) 0 CH	2500–5000 IU b.i.d. 0/200 (0) 1 non-fatal CH	RR 0, n.s.; Lower mortality despite >Rankin & BP	258 (87 to _270)	C
Low molecular weight heparins/heparinoids versus placebo							
Bath [29]	2765 (3048), any CI	SRMA of 11 studies, Complete	5.6%	2.7%	DVT: OR 0.27 (0.08–0.96); PE: OR 0.34 (0.17–0.69) BUT extracranial haemorrhage: OR 2.17 (1.10–4.28)	? OR mortality 1.05 (0.83–1.32); OR symptomatic CH 1.77 (0.95–3.31)	A
Turpie *et al.* [24]	75, CI < 7 days	RCT, Complete; fibrinogen uptake	7/25 (28)	IV/SC Orgaran 1000 IU b.i.d 2/50 (4)	RR 0.14 (0.03–0.64)	4 (2–18)	B
Prins [25]	60, CI < 3 days	RCT, Complete; fibrinogen uptake	15/30 (50)	Fragmin 2500 IU b.i.d 6/30 (20)	RR 0.4 (0.18–0.89)	3 (2–14)	B
Sandset [27]	92 (103), CI < 3 days	RCT, Complete; phlebography	17/50 (34) Proximal - 8/50 (16)	Fragmin o.d. 55–65 IU/kg/day 15/42 (36) Proximal - 5/42 (12)	RR any DVT 1.05 (0.6–1.84)	_ 58 (6 to _5)	B
Elias *et al.* [26]	30, CI (<?)	RCT, Complete; fibrinogen uptake	12/15	CY 222 15 000 IU/day 0/15	0 ($P = 0.000$)	1 (1–2)	C
Kay *et al.* [3]	312, CI < 2 days	RCT Complete; clinical	1/105 (1)	Fraxiparine 4100 IU o.d. or b.i.d 0/203 (0): o.d. 0/102, b.i.d. 1 × 0/101	0 (n.s.)	105 (36 to _110)	C
TOAST [28]	1281 (1174), IC < 1 day	RCT, Complete; clinical	10/635 (1.6)	IV Danaparoid (Orgaran) in adjusted dose (0.6–0.8 anti-Xa IU/ml) 2/646 (0.3)	OR DVT 3 months 0.20 (0.04–0.89), (D2 and D10 n.s.)	79 (43–485); BUT: severe CH in 11 vs. 3 (D2), 15 vs. 5 (D10: $P < 0.05$: OR 2.9), 16 vs. 8 (3 months)	C
Hommel [6]	767, CI < 1 day	RCT, Abstract; clinical	Any PE 14/250 (5.6)	Fraxiparine 86 IU/kg o.d. or b.i.d. Any PE 9/516 (1.7): o.d. 4/271 (1.5), b.i.d.5/245 (2)	RR any PE 0.31 (0.14–0.71)	26 (14–127); BUT ↑ in CH and haemorrhage in general with b.i.d. regimen	C

Table 1 (Continued)

Reference number	Patients	Trial design	DVT in controls – *n* (%)	DVT with intervention – *n* (%)	RR/OR[a] (95% confidence interval)	NNT[b]	Study quality
Low molecular weight heparins/heparinoids versus unfractionated heparin							
Dumas *et al*. [32]	179 (175), CI < 3 days	RCT, Complete; fibrinogen uptake	5000 b.i.d. 17/86 (19.8)	Orgaran 1250 IU o.d. 13/89 (14.6)	RR 0.758 (0.402–1.429)	19 (6 to _17)	A
Counsell [34]	705, CI < 14 days 5 studies	SRMA of 5 studies with prospective evaluation, Complete	65/291 (22)	55/414 (13) Fraxiparin, Enoxaparin or Orgaran	0.52 (0.56–0.79), BUT n.s. in worst scenario (losses etc.)	11 (7–31) PE, symptomatic DVT, bleedings – small numbers	A
Turpie *et al*. [31]	87, CI < 7 days	RCT, Complete; fibrinogen uptake	5000 b.i.d. 13/42 (31.0)	Orgaran 750 IU b.i.d. 4/45 (8.9)	RR 0.29 (0.10–0.81); n.s. for proximal DVT or bleeding	5 (3–17)	B
Hillbom [33]	148 (212), CI < 2 days	RCT, Complete; phlebography	5000 t.i.d. (25/72 (34.7%)	Enoxaparin 40 mg o.d. 15/76 (19.7%)	RR 0.57 (0.33–0.99)	7 (3–124)	B
Hossman [30]	30, CI < 1 day (?)	RCT, Abstract; fibrinogen uptake	5000 b.i.d.	Orgaran in 2 regimens: 750 or 1250 anti-Xa IU b.i.d.	4 DVT n.s.	Primary data for calculation not shown	C
Low molecular weight heparins/heparinoids versus aspirin							
Berge [56][c]	11 721, CI < 2 days, 4 studies	SRMA of 1 B (unpublished) and 3 C- quality studies, complete	Primary data for calculation not shown	Symptomatic DVT – OR 0.19 (0.07, 0.58), BUT n.s. ↓ symptomatic PE 0.85 (0.55–1.32) AND ↑ major extra-cranial - OR 1.94 (1.20–3.12) - and intracranial bleeding – OR 2.27 (1.49–3.46)		Primary data for calculation not shown	B
Berge [38]	449, CI with AF < 30 h	RCT, Complete	160 mg 5/244 (2.2)	Dalteparin 100 anti-Xa IU/kg o.d. 1/225 (0.4)	OR 0.20 (0.02–1.70), RR 0.22 (0.03–1.84)	62 (28 to _267)	C
Bath *et al*. [39]	1486 (1438), CI < 2 days	RCT, Complete; 2 doses of tinzaparin (100 and 175 anti-Xa IU/kg/day)	Aspirin 300 mg DVT 9/482 (1.8%), total VTE 13, CH 1;	Higher dose: DVT 0/954 (0%), total VTE 2, 7 CH;	Significant ↓ (DVT & total VTE) with higher dose, BUT also significant ↑ of CH; OR 0 for DVT	54 (33–152)	C
Leonardi-Bee [40]	1933, 2 studies HAEST, TAIST	SRMA of 2 C quality studies, Abstract	14/726 (1.9)	1/1179 (0)	Symptomatic DVT - OR 0.29 (0.12–0.66), BUT ↑ OR major extra-cranial bleeding 2.57 (1.01–6.52).	54 (35–121)	C

Table 1 (Continued)

Reference number	Patients	Trial design	DVT in controls – *n* (%)	DVT with intervention – *n* (%)	RR/OR[a] (95% confidence interval)	NNT[b]	Study quality
Aspirin							
Cameron [36]	112, CI < 3 days	RCT, Complete; daily thermography	Aspirin (300 mg t.i.d.) ± dipiridamol (100 mg t.i.d.) 22/112 (19.6%) in 10 days		Primary data for calculation of eventual differences between combination versus aspirin alone not shown		C
CAST [4]	20 665, CI < 2 days	RCT, Complete; clinical	20/10 320 (0.2%)	160 mg 12/10 335 (0.1%)	RR 0.60 (0.29–1.22)	1287 (541 to _3378)-	C
IST [5]	19 433, CI < 2 days	RCT, Complete; clinical	77/9715 (0.8%)	300 mg 57/9720 (0.6%)	RR n.s. 0.74 (0.53–1.04)	485 (228 to _3774)	C
Kelly [16]	102, CI < 7 days	Prospective observational, Complete	-	Aspirin + IPC VTE in 40% (85% asymptomatic); DVT 29%, PE 11%	-	-	C
Other drugs							
Mellbring [41]	50, CI < 2 days	RCT, Complete; fibrinogen uptake; IV Dextran 3 days	13/26 (50)	13/24 (54)	RR 1.06 (0.64–1.84)	_ 24 (4 to _3)	B
Ginsberg *et al.* [42]	88 (103), subacute CI	RCT, Complete; ultrasound & phlebography; Warfarin 2 mg/day	7/41 (17) Proximal 5 (12)	3/47 (6.3) Proximal 1 (2)	RR any DVT 0.39 (0.13–1.37); RR proximal DVT 0.17 (0.01–1.4)	9 (4 to _36) for any DVT	B
Harvey [37]	1506, CI & CH rehabilitation (1030 CI)	Observational, complete; Multiple regimens[d]	VTE rates were reduced with all regimens, but only therapeutic anticoagulation reached statistical significance (low-dose heparin also after adjustments)		After adjustments: OR therapeutic anticoagulation: 0.37 (0.15–0.88), heparin prophylaxis OR 0.48 (0.23–0.98)	Primary data not shown for calculation	C

Notes: 3 articles not found: refs. [57–59]. DVT, deep venous thrombosis; IPC, intermittent pneumatic compression; OR, odds ratio; RR, relative risk; NNT, number needed to treat; RCT, randomized clinical trial; CH, cerebral haemorrhage; CI, cerebral infarct; PE, pulmonary embolism; SRMA, systematic review with meta-analysis; BP, blood pressure; IV, intravenous; sc, subcutaneous; IU, International Units; n.s., not significant; Complete, published in complete form; Abstract, published in abstract form; *n*, number of patients studied; _ (underline) refers to negative numbers.
[a]Relative risk if otherwise not specified in the original article. No important differences between RR and OR calculations were found in the re-evaluation of data from all articles; [b]Number needed to treat – were when needed calculated by the authors based on the original data.; [c]The Cochrane review from Berge and Sandercock first appeared in 2001; it also includes the comparison between anticoagulants plus aspirin versus aspirin alone in 9720 patients (see Discussion); [d]Therapeutic anticoagulation with either oral Warfarin or parenteral anticoagulation with enoxaparine ≥60 mg b.i.d.; low-dose prophylactic use of anticoagulants (subcutaneous unfractionated heparin 5000 IU b.i.d. or t.i.d., enoxaparine 30 mg b.i.d. or 40 mg o.d., ardeparin 50 IU/kg/day); aspirin and other antiplatelet drugs; and combinations of low-dose prophylactic anticoagulants and antiplatelet drugs.

been suggested by a single small study. The association of aspirin and IPC also seems to be ineffective. Its usefulness in patients with CH has been suggested and merits more study.

Parenteral anticoagulants

We found five level A studies including three systematic reviews and two clinical trials. We also found 18 additional clinical studies with one or more important limitations. Studies evaluating UFH usually involved subcutaneous (SC) administration every 12 or 8 h. LMWH/heparinoids studies involved administration of fraxiparin, tinzaparin, dalteparin, enoxaparin and the heparinoid Orgaran (ORG 10172) once or twice daily.

Unfractionated heparin versus placebo

Seven studies directly compared UFH and placebo in CI or CH patients [5,17–22]. There is also a systematic review evaluating most of these studies [2].

All clinical studies (most with small numbers and non-significant differences found) showed trends for decreased incidence of VTE in acute CI patients, with a dramatic reduction of DVT with UFH 5000 IU t.i.d. – incidence of 22% in treated patients versus 73% in the placebo group – and on the combined rate for DVT, PE and death clearly demonstrated in the largest published study to date [20]. In general, the effect on DVT is more clearly shown than that on PE. This was confirmed in the published meta-analysis [2].

Two trials evaluated patients with CH [21,22]. All have major limitations. One study compared UFH (5000 IU every 8 h) initiated 4 or 10 days after symptom onset in only 46 patients [21]. There were no significant differences in the rates of DVT (1 patients vs. 2 patients), PE (5 vs. 9) or rebleeding (1 vs. 2) as evaluated in the 10th day. The same group later reported similar findings with no bleeding complications in patients receiving UFH from the second day on [23]. The other study (only published in abstract form) retrospectively evaluated clinical DVT in 458 patients [22]. UFH (2500–5000 IU b.i.d.) was given to 200 patients (no details on the time elapsed from symptom to treatment onset). Only one patient (in the control group) developed DVT in the whole series and rebleeding was also exceptional (one case in the treatment group). Taken together, these studies suggest the risk of bleeding is not prohibitively high, but no significant effect on VTE rates was detected.

Low molecular weight heparins and heparinoids versus placebo

Seven studies evaluated LMHW (essentially fraxiparin) or the heparinoid Orgaran against placebo [3,6,24–28]. There is an additional meta-analysis published and evaluating many aspects of the use of LMWH/heparinoids in acute stroke patients [29]. All but one study [27] revealed trends for reduced rates of DVT or PE in treated patients. The low number of haemiparetic patients included in studies systematically screening for DVT limits their power for final conclusions.

Studies with no prospective evaluation support the idea of a reduction of PE and DVT in patients with reduced mobility. However, these relatively large studies point against the use of LMWH/heparinoids in unselected populations of CI patients because of the increased risk of systemic and intracranial bleeding. The same conclusion is derived from the published meta-analysis [29].

Low molecular weight heparins/heparinoids versus unfractionated heparin

Four clinical trials comparing LMWH or heparinoids (Orgaran) with UFH [30–33] and one meta-analysis of interest [34] were found. There is a non-significant trend suggesting increased efficacy of LMWH/heparinoids for the prevention of DVT in CI patients with limited mobility. However, the number of deaths, PE episodes and bleeding episodes are low and do not allow for a firm conclusion on a differential impact of the two classes of drugs over these more important clinical endpoints. There is no proof that the effect over DVT in unselected patients with acute CI is translated into reduced mortality or PE events and in fact there is a significant increase of extracranial bleeding episodes and a tendency towards increased risk of intracranial bleedings and mortality [34].

Aspirin

Aspirin is the only drug whose routine use has been confirmed to reduce mortality and early (14 days) stroke recurrence rates in unselected patients with acute CI [4,5,35]. However, its use is not usually considered a reasonable single strategy for the prevention of VTE in this setting. Four studies – two large studies without prospective evaluation [4,5], one with insufficient data reported [36] and one observational [16] – suggest a modest beneficial effect on DVT and PE rates, with very large numbers of patients needed to treat in order to prevent a single episode. The observational study suggested that aspirin plus IPC is not effective [16]. Aspirin also seemed ineffective in a large retrospective study of patients undergoing rehabilitation for stroke [37].

We performed a pooled analysis of data from two large prospective trials evaluating the effects of aspirin in patients with acute CI (Fig. 1) [4,5]. No prospective evaluation for VTE was made in these studies. PE events were clinically diagnosed, with no formal predefinition of diagnostic criteria; and physicians were

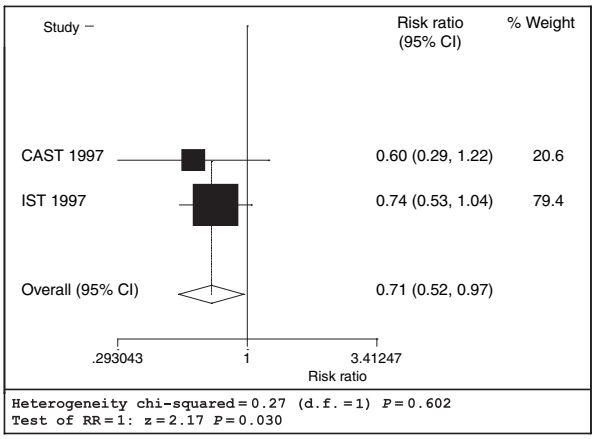

Figure 1 Meta-analysis of data from two large trials evaluating aspirin use in 40090 acute stroke patients [4,5]. The incidence of clinically diagnosed pulmonary embolism was compared in patients receiving aspirin (160 to 300 mg/day) early in the clinical course (up to 48 h) or not receiving it. There is a highly significant reduction of events in the group using aspirin. Heterogeneity chi-squared = 0.27 (d.f. = 1) $P = 0.602$. Test of relative risk = 1: $z = 2.17$, $P = 0.030$.

aware of the allocation of patients either to aspirin or no treatment. Despite these limitations and the relatively low number of reported events, the results of the pooled analysis strongly suggest a mild protective effect of aspirin against VTE or at least PE, its most severe manifestation. There was a significant reduction of the frequency of any apparent PE [69/20055 (3.4%) vs. 97 (4.8%) in 20035 controls – RR 0.71 (0.52–0.97)] and a non-significant reduction of fatal PE [39 patients (1.9%) vs. 56 controls (2.8%) – RR 0.70 (0.46–1.05)]. However, it should be remembered that the beneficial effects of aspirin on DVT prophylaxis were counterbalanced by a significant rise in the rate of major extracranial haemorrhages (fatal or requiring transfusion) – RR 1.69 (1.35–2.13) and a non-significant increase in the rate of CH – RR 1.22 (0.95–1.56).

Low molecular weight heparins versus aspirin

Two studies directly comparing aspirin and LMWH – tinzaparin and dalteparin – in acute CI patients and one meta-analysis were found [38–40]. One study specifically evaluated patients with presumed cardio-embolic strokes related to atrial fibrillation [38]. The primary aim of these studies was to evaluate mortality and stroke recurrence (non-significant differences in both) and VTE was not evaluated systematically. However, analysis of pooled data from these studies clearly indicates a reduction of symptomatic DVT episodes and a rise in major extracranial bleeding complications in patients receiving LMWH [40].

Other antithrombotic drugs

We found three additional studies of interest. One small trial did not show significant differences between CI patients receiving dextran and controls – 54% vs. 50% [41]. A non-significant reduction of fatal PE was found – 1 episodes vs. 3 episodes.

The other controlled study was well-conducted but only included patients in the rehabilitation phase following CI and was flawed by a significant number of losses to final evaluation [42]. Use of Warfarin in fixed low dose (2 mg) was associated with a non-significant reduction of any DVT, proximal and clinically detected DVT but not with any bleeding complication. The authors suggest that this fixed low dose is almost never associated with a prothrombin time/international normalized rate (INR) above 2.0 and that a single INR measurement at the end of the first week of treatment is enough to warrant safety with this regimen.

Moreover, in the rehabilitation setting, a large retrospective study suggested that patients with reduced mobility secondary to CI or CH and who underwent oral or parenteral full anticoagulation exhibit a lower rate of VTE episodes when compared with those receiving SC low-dose LMWH or antiplatelet drugs [37]. Although the results of this non-randomized study should be taken with reservations, an increased risk of bleeding complications could not be detected with any of the multiple antithrombotic strategies studied.

Discussion

In patients with reduced mobility, stroke is strongly associated with an increased risk of VTE. Different series prospectively evaluating VTE with ultrasound or fibrinogen scan reveal an incidence between 15% and 80%, with higher figures in the side of paralysis and in severe cases [12–14,16–18,20,36,43,44]. Studies evaluating only clinically evident DVT find much lower numbers (0.2–10%) [19,22,37,45] that are clear underestimates of the true incidence. Even patients under different forms of prophylaxis may be at risk (0–46%) [12,15,16,36]. A recent study using direct magnetic resonance imaging of thrombus confirmed a high incidence of any VTE (40%), proximal DVT (18%) and PE (12%) in unselected patients with acute CI [16]. Most cases were asymptomatic or unrecognized: DVT and PE were recognized in only 1% and 3% of patients respectively, with many other clinically evident cases not recognized by the attending physicians.

The central role of reduced mobility in the pathogenesis of DVT is highlighted by consistently higher rates in the paretic limbs [33,46]. Sioson *et al.* [46] reported 19 DVT events in the paretic limb, nine bilateral events and four contralateral in 32 patients

Table 2 Factors increasing the risk of venous thrombo-embolism (VTE) after stroke. The strength of evidence may vary (see text)

Neurological factors	Systemic conditions
Any reduction in mobility[a]	Previous VTE, post-phlebitic syndrome
Paretic lower limb	Increasing age
Dense haemiplegia	Known disseminated malignancy
Increasing stroke severity	Congestive heart failure
Haemorrhagic stroke	Atrial fibrillation
Increasing delay between stroke and hospital admission or prophylaxis onset	Morbid obesity
Cryptogenic stroke	Pro-thrombotic states: acquired (e.g. hormone treatment) or genetic
	Dehydration during hospitalization

[a]Almost all deep venous thrombosis episodes occur in patients unable to walk independently. However, the risk of pulmonary embolism may persist well after patients regain walking ability.

prospectively followed. Additional risk factors have now been identified and include severity of the illness, age, dehydration along the hospitalization period, increasing time between stroke onset and the introduction of preventive measures, haemorrhagic stroke, cryptogenic ischaemic stroke [33,44,47–52]. Current guidelines and reviews also mention other factors including previous VTE, morbid obesity and known pro-thrombotic conditions including contraceptive use (Table 2) [7–9].

We confirmed that there is strong scientific evidence that low-dose UFH and LMWH are partially effective in the prevention of DVT after acute CI. In most prospective studies with systematic search for VTE episodes, between 2 and 10 patients need to be treated for one episode being prevented (Table 3). The dose regimen used in diverse studies is remarkably variable. UFH is usually given in two or three daily SC doses of 5000 IU, with no obvious differences in efficacy or bleeding risk. Many LMWH formulae are available with most studies evaluating 60–175 IU/kg/day SC injections. A typical regimen would consist in 55–100 IU/kg/day given in one or two SC injections. Higher doses tend to be associated with more marked preventive effect but also an increased risk of bleeding complications.

However, the associated risks seem to overcome any potential benefit when unselected CI patients are treated. Patients without reduced mobility – usually related to lower limb paresis – should not be routinely given low-dose anticoagulants, as their VTE risk is probably very low and they may still suffer bleeding complications.

Mechanical methods, although favoured by some, are not effective in acute stroke patients with IC and reduced mobility [53]. Even in patients receiving low-dose aspirin, IPC is associated with a high frequency of VTE [16]. The use of IPC devices in patients with clear contra-indications for anticoagulant use – increased bleeding risk in CI patients, patients with CH or subarachnoid haemorrhage – has not been adequately studied and its recommendation is based mainly in expert opinion. There is now some evidence that IPC plus elastic stockings may be more effective in the reduction of asymptomatic DVT (in both distal and proximal veins) in acute CH patients than the use of elastic stockings only [11]. The relevance of these findings is still unsettled: in the same study, for instance, only one patient in each group developed symptomatic DVT,

Table 3 Approximate number of patients needed to treat (NNT) for the prevention of one episode of venous thrombo-embolism following stroke. The usual dose range is also shown

Intervention	Usual regimen	Number needed to treat[a]
UFH	SC – 5000 IU two or three times a day	2–10
LMWH	SC – 55–100 IU/kg/day (one or two injections)	1–4[b]
Warfarin[c]	PO – fixed low-dose (2 mg) or full anticoagulation (target-INR: 2–3)	9
Mechanical methods[d]	Intermittent compression whilst in bed	9–16
Aspirin	PO – 160–300 mg/day	> 300 (> 1000 for PE)

UFH, unfractionated heparin; LMWH, low molecular weight heparin; SC, subcutaneous route; PO, oral route; PE, pulmonary embolism; INR, international normalized rate.
[a]Numbers found in prospective studies with systematic evaluation of symptomatic or VTE; protective effect better documented for deep venous thrombosis than for pulmonary embolism; [b]Studies comparing LMWH with placebo; LMWH may be somewhat more efficient than UFH (see text); [c]NNT refers to low-dose fixed regimen of 2 mg/day in the rehabilitation setting (one single study); [d]Weak evidence for any method: graded elastic stockings usually considered less efficient than intermittent pneumatic compression devices.

with no episode of PE diagnosed. We suggest that even in these patients the risk of bleeding complication should be frequently reassessed along the disease course and low-dose anticoagulants started as soon as it comes down to acceptable levels. Given the low risks associated with pneumatic mechanical devices (contra-indications: peripheral artery disease, skin lesions), they might be used in patients considered to have a high risk of bleeding complications with anticoagulants, such as those with active peptic disease or bleeding or those with primary (non-traumatic) CH; or in association with other methods in patients considered to have an exceptionally high risk of VTE [54]. However, even in patients that cannot be initially treated with low-dose anticoagulants the bleeding risks and the possibility of instituting them should be frequently reassessed along the hospital stay.

The use of low-dose UFH after CH was not associated with significant bleeding risk in low quality clinical series [21–23]. However, the impact over VTE frequency has been non-significant with very low rates detected in treated and control groups. The low incidence of VTE in these trials is not in perfect agreement with figures from other studies. Patients with CH may in fact be at higher risk for DVT than CI patients. A retrospective cohort study revealed a RR for clinical DVT after CH of 4.0 (1.8% vs. 0.4%) [50]. The RR for PE was 3.5 (0.4% vs. 0.1%). Data comparing patients with CH included in the IST with CI patients also suggest an increased risk of clinical PE – 1.3% vs. 0.7% ($P = 0.06$) – even after controlling for severity of the disease [48]. There is clearly a need for additional and well-designed trials in this population.

Unlike most reviews, we also evaluated the possible role of aspirin as a VTE preventive strategy. We found a modest preventive effect of aspirin over clinical or fatal PE when pooled data from 40 090 patients included in IST and CAST were evaluated. Although the quality of the data is necessarily poor as neither study prospectively evaluated PE or DVT and physicians and patients were not blinded to treatment allocation, aspirin use was associated with a significant reduction of the frequency of any apparent PE and fatal PE. However, the clinical relevance of these findings is questionable. This can be seen when number needed to treat are calculated – 714 (376–6946) for any PE and >1000 for fatal PE. Moreover, aspirin plus IPC seemed to be ineffective in a single observational study [16]. The possible harm (increase in severe extracranial haemorrhages and possibly CH) associated with aspirin use should also be taken account when deciding to use the drug for VTE prophylaxis or reduction of early recurrence.

The present systematic review includes a number of original features including a thorough review of the Brazilian literature on the subject and the search of databases usually absent in other reports. The review also includes studies only published in abstract form. Even recognizing that some of these studies may have non-evident methodological problems, as they were not submitted in complete form to peer reviewed scientific magazines, the inclusion of these studies is considered important to secure the completeness of the search [55].

Some particulars of the present study should be acknowledged. No unpublished study was included. Pooled analysis of data was only made for the use of aspirin. We believe meta-analyses should only be performed when there is adequate comparability of the trials [55] and this was not the case with many questions here addressed. We did not exclude studies with no prospective evaluation of VTE episodes and preferred instead to classify the studies as to their general scientific validity. Lastly, we did not evaluate potential differences between diverse formulations of LMWH or between them and heparinoids.

In conclusion, thrombo-prophylaxis following acute stroke seems far from ideal. A number of important questions remain unsolved and open to research. What is the role of IPC in association with antithrombotics – aspirin or low-dose anticoagulants? What are the risks and potential benefits with the association of aspirin and low-dose anticoagulants? When (if ever) should low-dose UFH or LMWH be safely initiated in CH patients? Additional unanswered questions pertain to the management of CI patients with very high risk for development of VTE (under contraceptive use or with marked obesity, congestive heart failure, disseminated malignancy, previous VTE) or of bleeding (e.g. immediately following administration of thrombolytics); the duration of antithrombotic prophylaxis in patients with reduced mobility (should they still receive the drugs after discharge? What degree of recovered mobility is associated with a decreased risk?); and the role of diverse regimens of parenteral or oral anticoagulants in the rehabilitation phase. All these questions can only be answered by future adequately planned and sufficiently large clinical trials clearly distinguishing the effects over asymptomatic calf vein DVT and outcomes of greater clinical relevance such as PE and proximal and symptomatic DVT.

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